

PROBLEM STATEMENT #49 | Developer Tools & IT Operations

Webhook Ingestion & Retry Mechanism Hub

1. Problem Context & Overview

A resilient webhook gateway receiving third-party events, logging request payloads, delivering to internal services, and managing dead-letter queues with exponential backoff retries.

Target Stakeholders / Actors: *Integration Developer, System Operator*

2. Sample Functional Requirement (FR) Guideline

Sample Requirement: FR-001 [Priority: High]

Description: The system shall ingest inbound webhooks, forward payloads to target endpoints, and automatically retry failed deliveries using exponential backoff (1s, 2s, 4s, 8s).

Sample Acceptance Criteria: Pass: Failed delivery successfully retried and logged; Fail: Dropped webhook without dead-letter persistence.

3. Sample Non-Functional Requirement (NFR) Guideline

Sample Requirement: NFR-001 [Type: Performance & Security]

Description: The ingestion endpoint shall buffer inbound webhooks with 99.999% reliability and throughput up to 2,500 requests/second.

Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]

- **Complete Requirements Table:**

- >> Exactly 5 Functional Requirements (FR-001 to FR-005)
- >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

- **UML Use-Case Diagram:**

- >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.

- **Use-Case Flow Specification:**

- >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
- >> One Alternate Flow.